TELCO MicroServices (µS) Environment set-up

Welcome

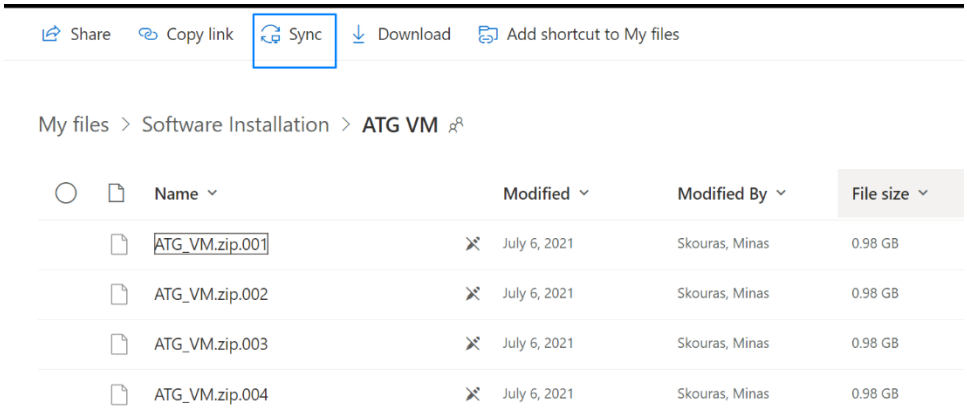
Welcome on the TELCO MicroServices team (hereon forward µS). We are so glad you are joining us. If you are reading this document, it means that you got assigned to our team either internally or externally. If you have been internally transferred some of the steps will already be familiar and you could safely skip them. However, we do encourage you to carefully skim through them just to make sure your environment is aligned. If you have been externally transferred, please, carefully pass through every step in the order presented.

We know that reading documentation is not a developer's favorite part of the day, so we tried to make this document as clear as possible. So, get your cup of coffee, take a deep breath, and prepare for some dreaded, but essential, environment set up.

See something in the document that needs attention? Shoot an email to xxx.xxx@xxx.com and we will update the document as needed.

Step 1 (Setup the VM)

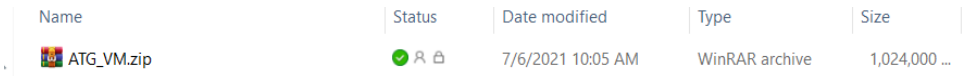
The first thing every member of the TELCO team in xxx needs to do is download the VM zip files. Due to size and depending on your internet speed this might take a while. Good news is you do not have to manually download each zip file. You can select to sync the content of the folder on your local machine.



You can safely let your pc do the rest. Even if the screen shuts off, the syncing will **not** stop!

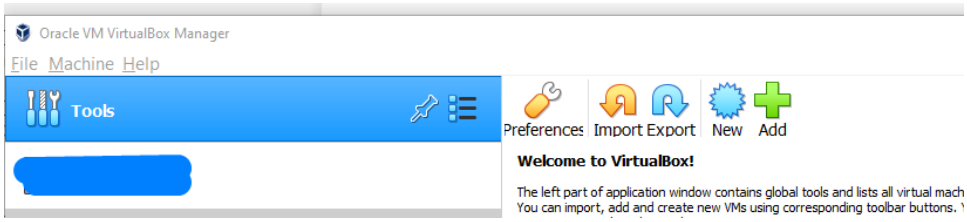
In the meantime you can download VirtualBox from here. Once downloaded, run the .exe to install the software.

Once the VM files are all downloaded and ready you will find a .zip file as shown below.

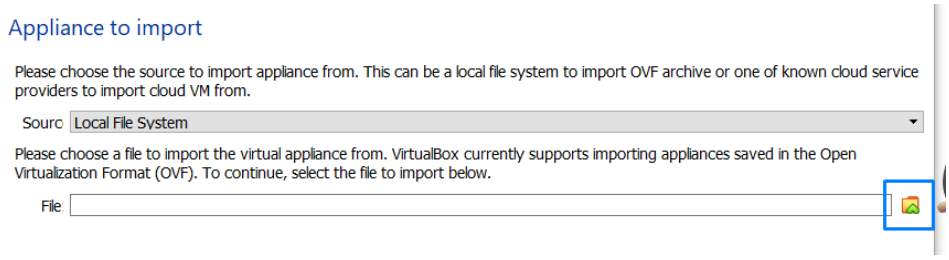


Create a new folder inside your user's page (e.g. C:\Users\firstname.lastname). You can name it as you desire but we suggest something VM/Virtual Box related, ATG_VM is a great name. Once folder is created, extract the .zip file into that directory. This will extract a .ova file of about 44GB. This is the ATG_VM and you need to import it on VirtualBox to create the machine.

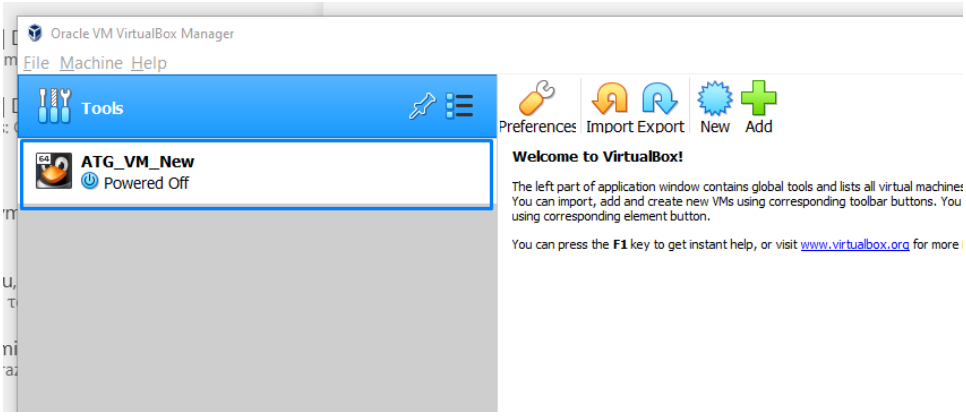
Open up VirtualBox. You will be greeted by this screen.



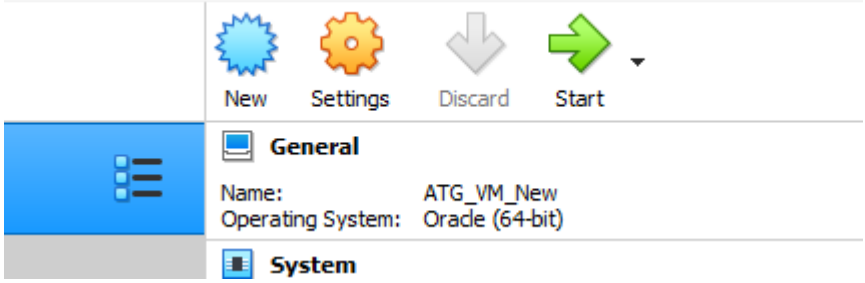
Click on import and browse to the .ova file you extracted a few minutes ago.



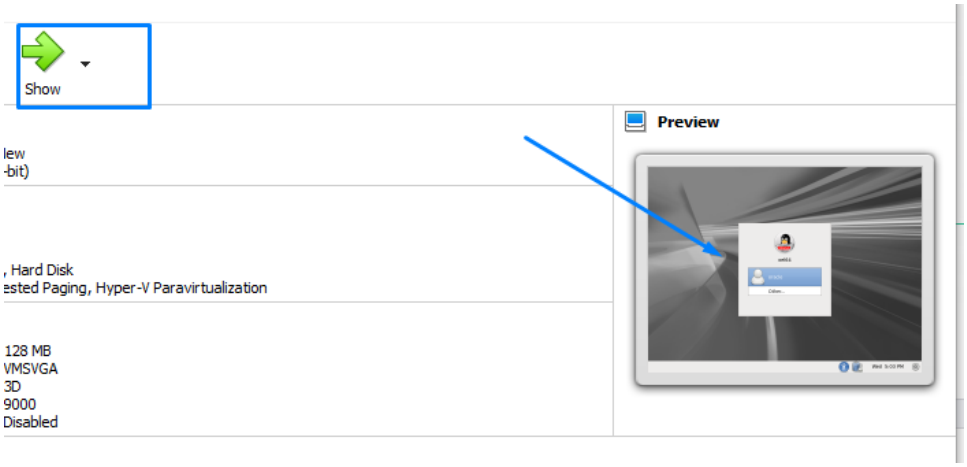
Once you find it, click next and start importing the .ova file. It comes with some pre-defined information and properties. Do not modify any of them. The import is bound to take a few minutes. Once completed, you should be able to see the new machine on the VirtualBox starting screen.



By clicking on it you will be able to see the machine's details and of course run it. **Always** start the machine with a *Headless start* and choose to *see* the machine when ready.

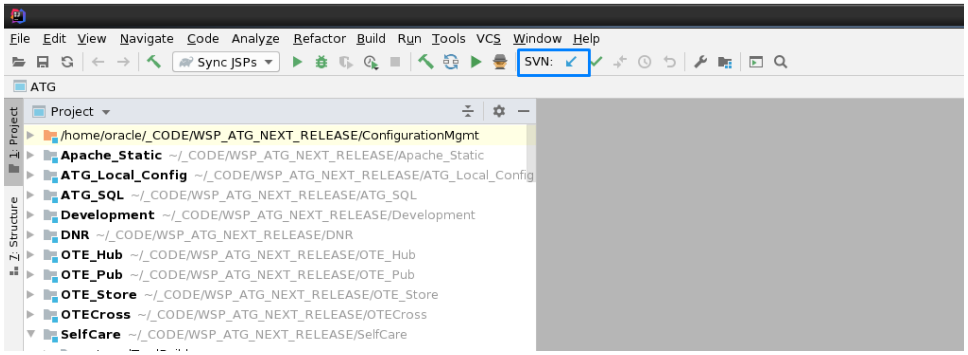


Do not click on start. Click on the black arrow next to it. This will show a dropdown menu. Click headless start

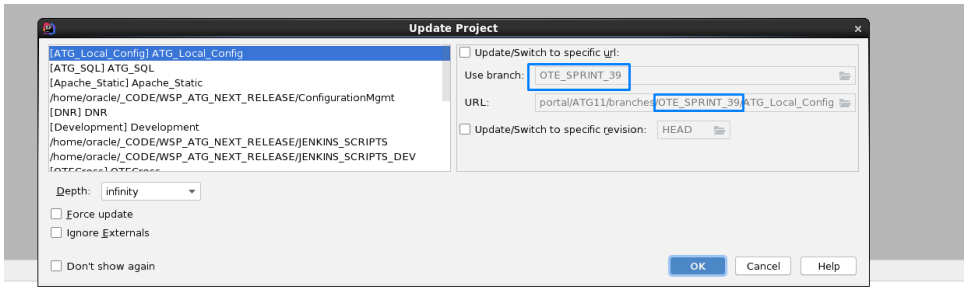


Once you see the login page on the preview window, click on show. Use password "oracle"

The machine is already set up. You do not need to install anything more. You can browse the ATG code by opening up IntelliJ. However, the code is a bit out of date, in other words you need to pull the latest code changes. Right now, the code base should be on sprint XX. You can use the SVN GUI on IntelliJ to quickly do so. Click on the blue arrow on the top of the screen.



This will open this window.



In your case, you should see OTE_SPRINT_XX. The screenshot above shows what you should be able to see after you pull the changes. To pull the changes, click on Update/Switch to specific URL and change OTE_SPRINT_XX on the URL property to OTE_SPRINT_XX. Repeat that for all projects on the left side list except XXX. Once the URL is updated for all projects, click the blue OK button; SVN will automatically pull all changes for all projects.

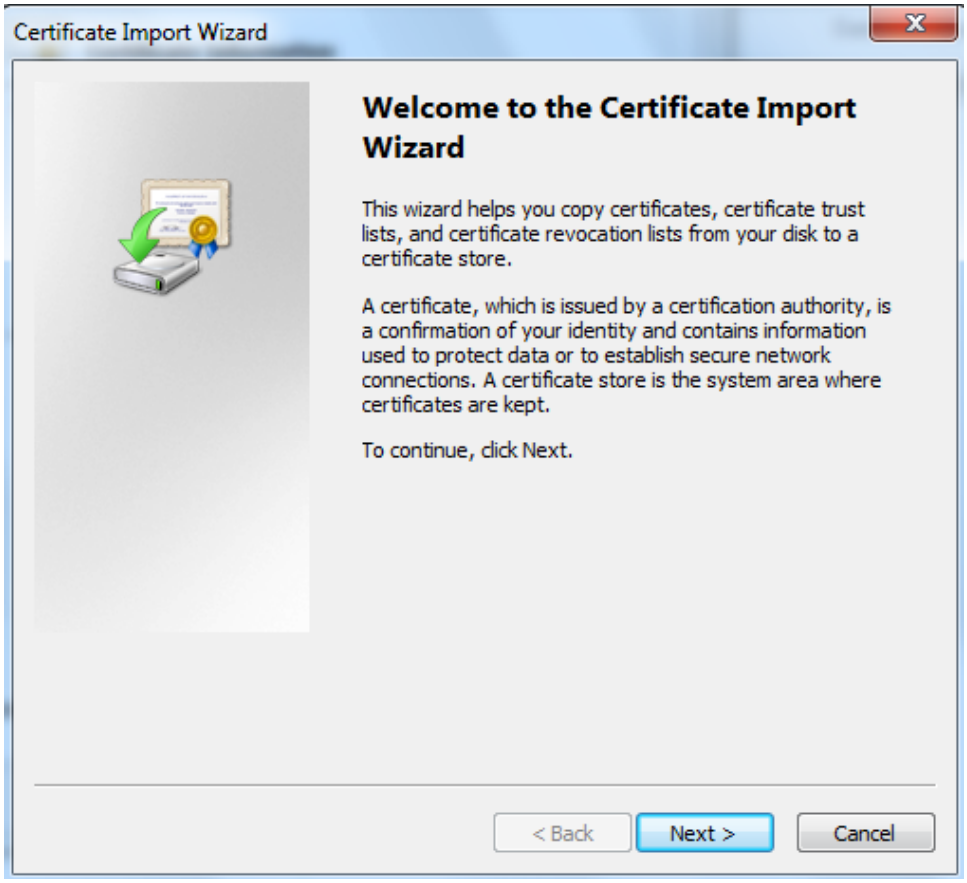
Step 2 (VPN)

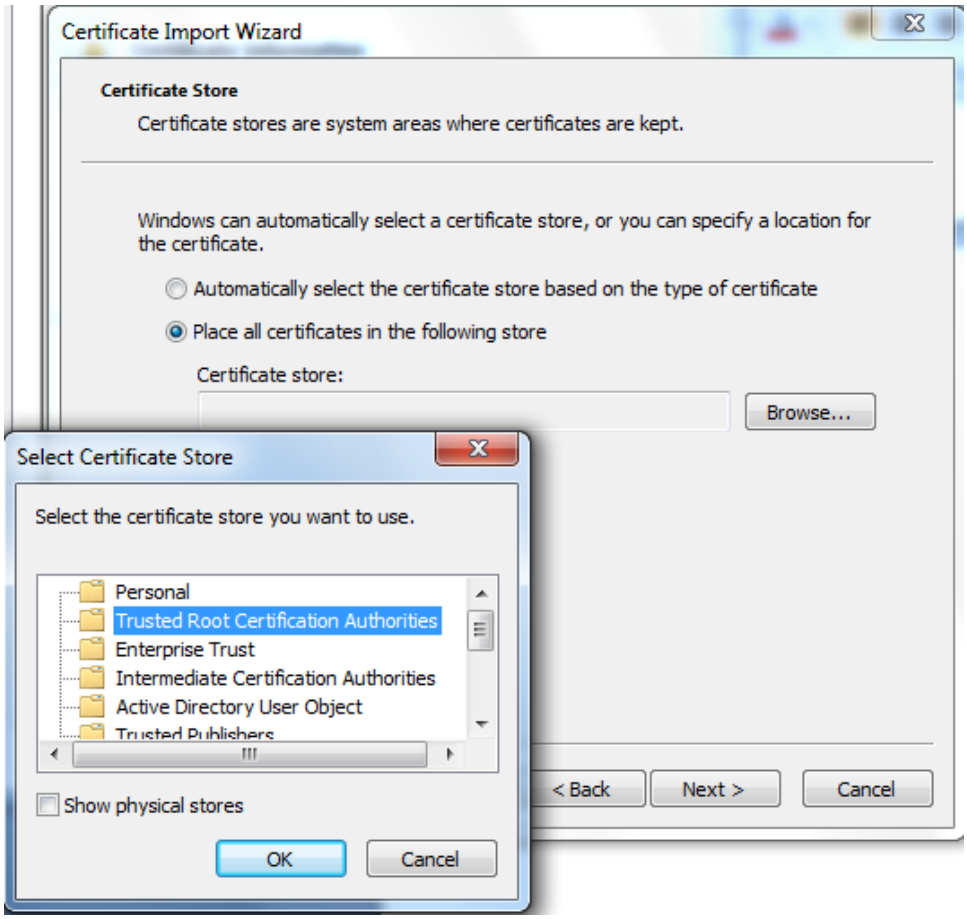
To access anything client (TELCO) related we must have a VPN connection established. For that reason, we use a specific VPN client and certificates provided by the client.

Start off by downloading all certificates. Place them in a folder as you please for later use. Additionally, you will need one personal certificate. To get said certificate you should contact one of the

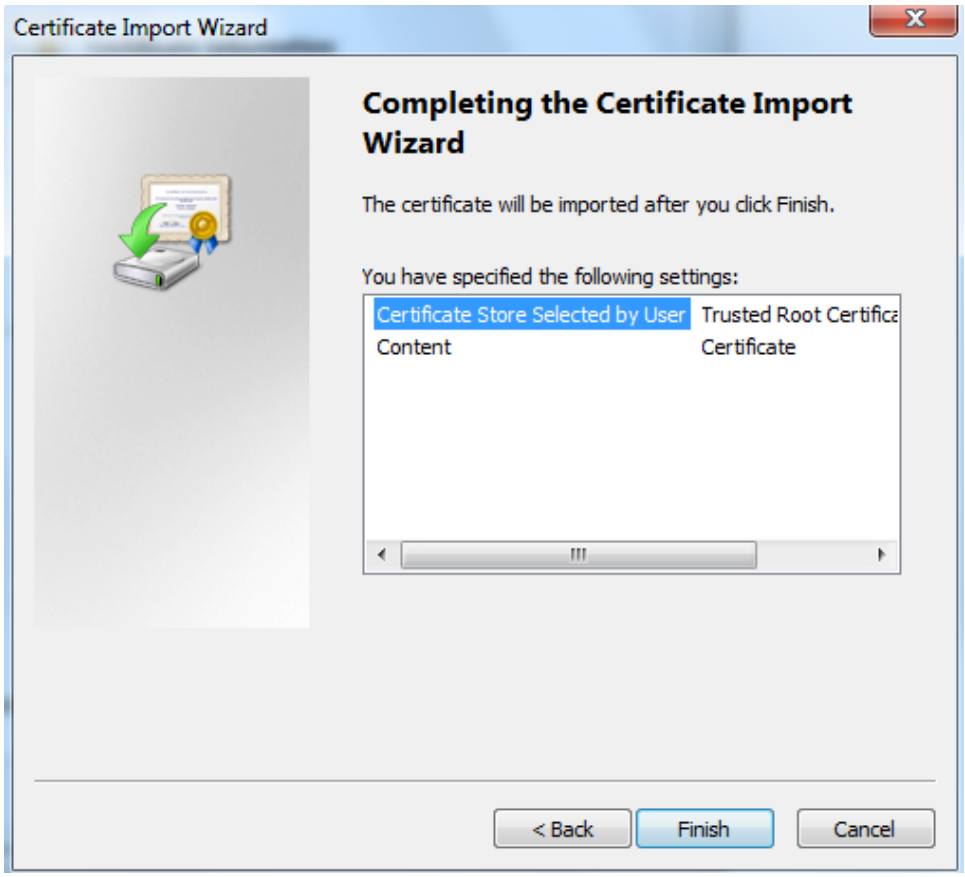
managers (XXX, XXX, XXX) for them to contact TELCO so they can create an IDM account. With that account creation you receive your personal certificate and password for VPN. Once you have the personal .crt you can proceed with the installations.

Start with *XXX.cer.*

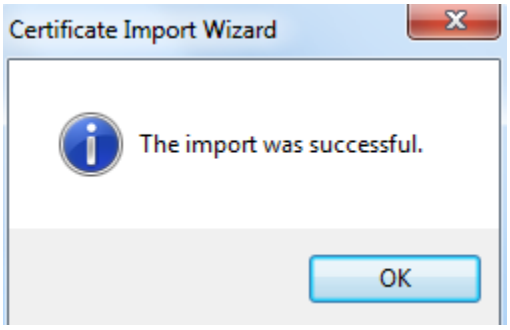



Certificate Import Wizard
Certificate Store
Certificate stores are system areas where certificates are kept.
Windows can automatically select a certificate store, or you can specify a location for the certificate.
Automatically select the certificate store based on the type of certificate
Place all certificates in the following store
Certificate store:
Browse...
< Back
Next >
Cancel
Select Certificate Store
Select the certificate store you want to use.
Personal
Trusted Root Certification Authorities
Enterprise Trust
Intermediate Certification Authorities
Active Directory User Object
Show physical stores
OK
Cancel

Click next.

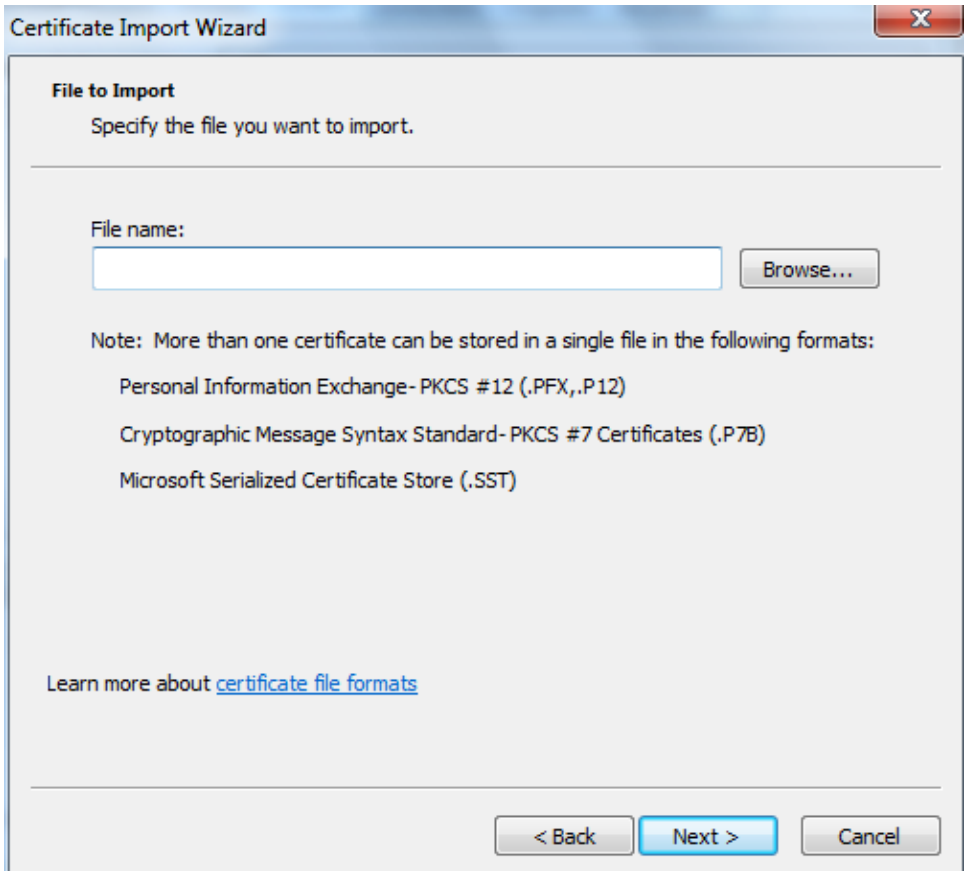


Click Finish.

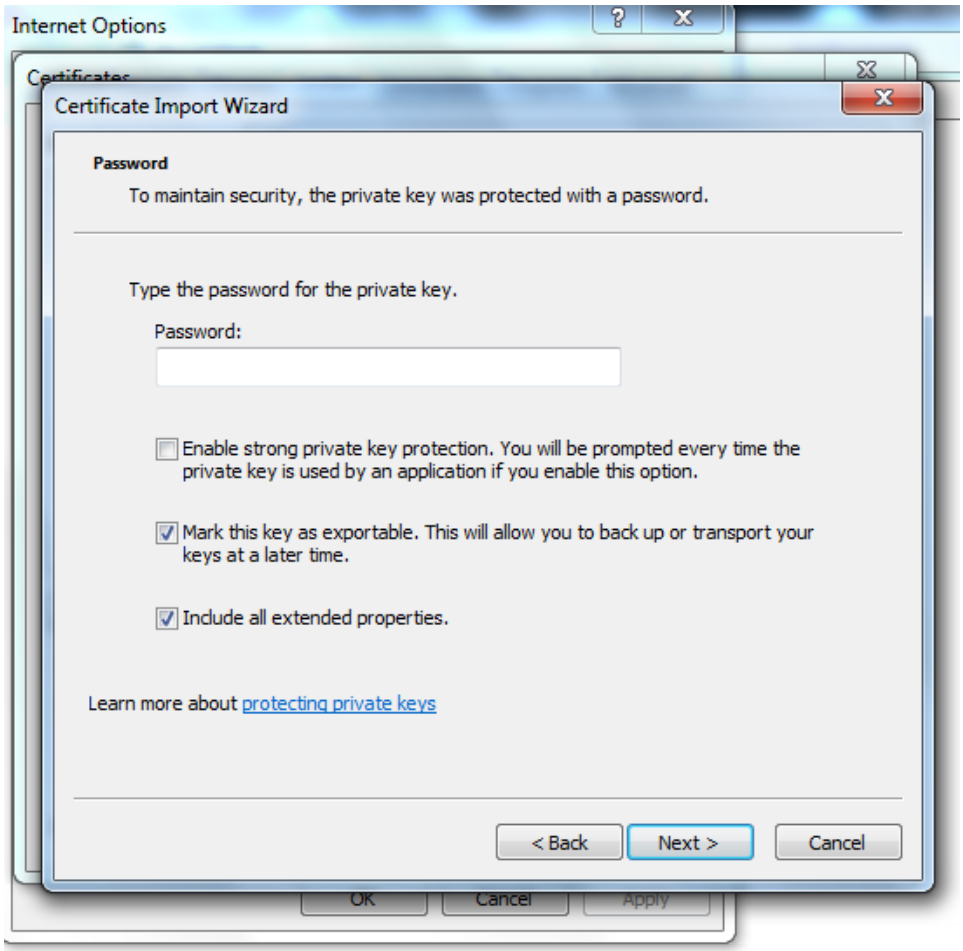


Next, following the exact same process import XXX.cer. After that, proceed with the personal certificate.

Internet Options
Certificates
Certificate Import Wizard
Welcome to the Certificate Import Wizard
This wizard helps you copy certificates, certificate trust lists, and certificate revocation lists from your disk to a certificate store.
A certificate, which is issued by a certification authority, is a confirmation of your identity and contains information used to protect data or to establish secure network connections. A certificate store is the system area where certificates are kept.
To continue, click Next.
< Back
Next >
Cancel
OK
Cancel
Apply

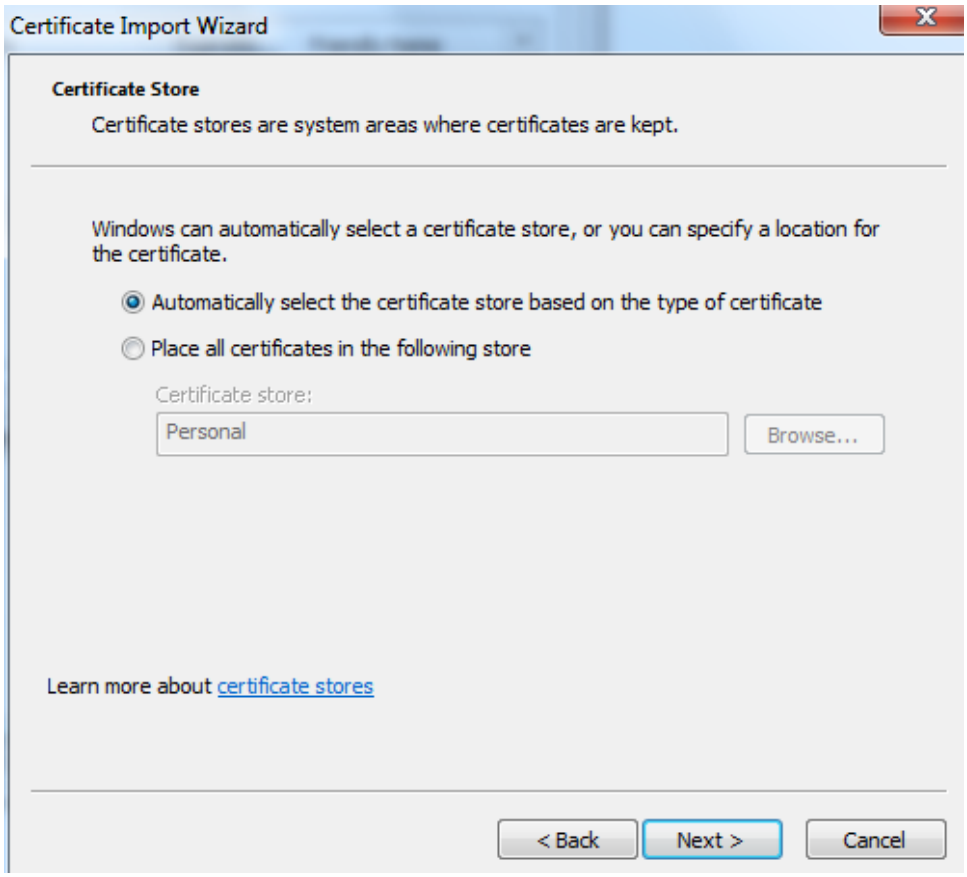


Open the file browser and select your personal certificate.



The password is the full name of the file without the extension. For example, if the certificate is named x.xxx@xxx.com.pfx then the password is x.xxx@xxx.com

Make sure that Mark this key as exportable and include all extended properties boxes are checked.


Certificate Import Wizard
Certificate Store
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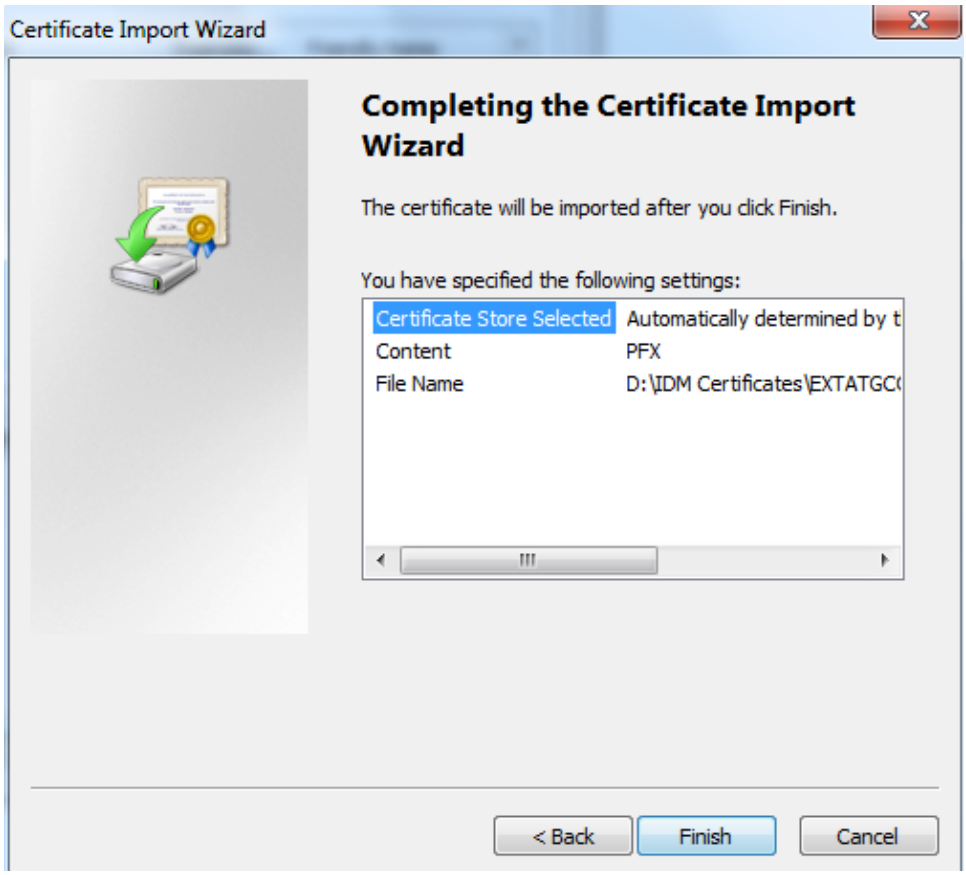
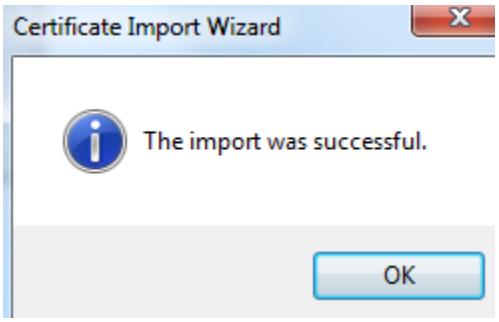
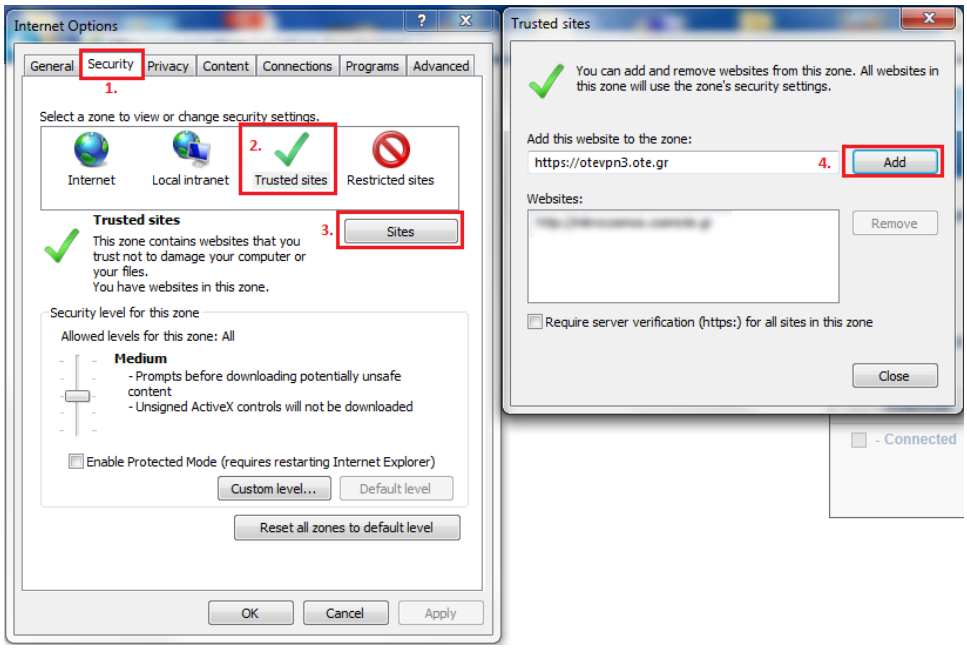
Windows can automatically select a certificate store, or you can specify a location for the certificate.
Automatically select the certificate store based on the type of certificate
Place all certificates in the following store
Certificate store:
Personal
Browse...
Learn more about certificate stores
< Back
Next >
Cancel





Once these three certificates are imported successfully you can move on to downloading the VPN client. To access it you need to access an OTE website that provides the link. To access that page, you need to add https://xxx.xxx.gr to the Internet Explorer trusted websites. Yes, Internet Explorer. Bear with us.



Once added, navigate to that link. You might be prompted to accept the certificate. Press OK. Then you will be asked for a password. The password for the VPN connection, now for this website but for the client later also will be communicated to you via SMS from TELCO.

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You will be greeted by a page that allows you to download Cisco AnyConnect. Download it and follow the installation wizard.

Once the installation is completed, startup AnyConnect and add otevpn3.ote.gr if not already present. Click connect and enter the password mentioned above.

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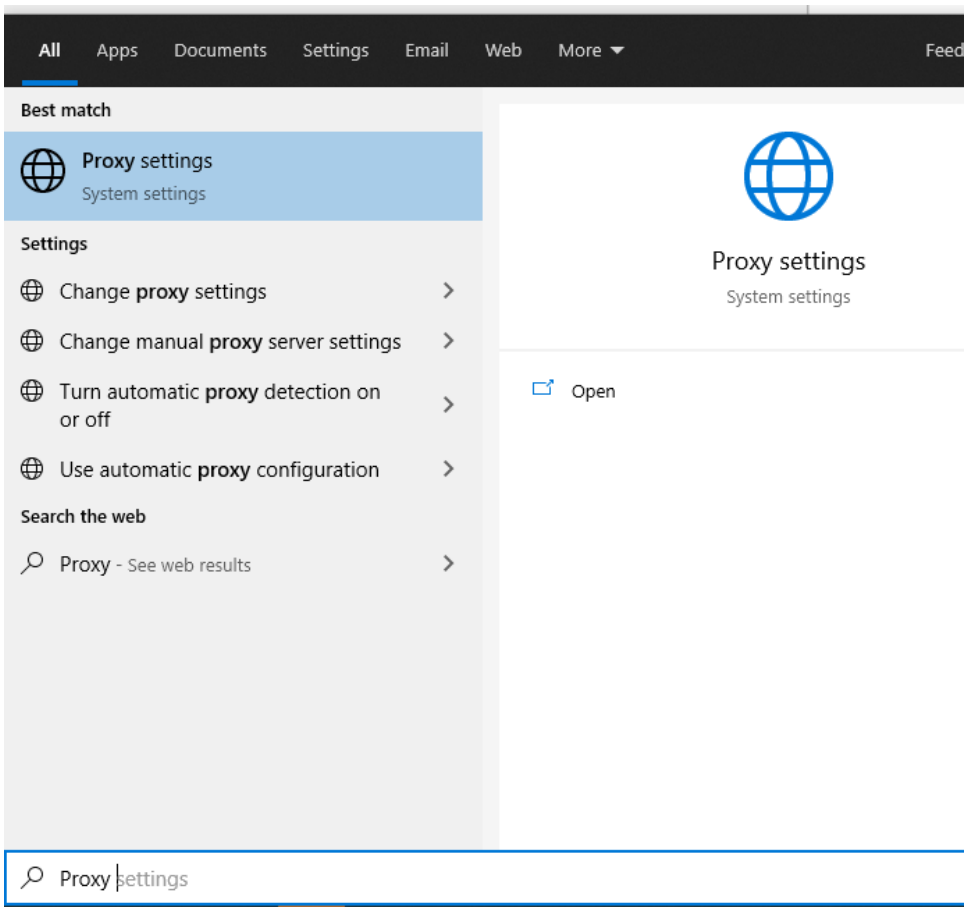
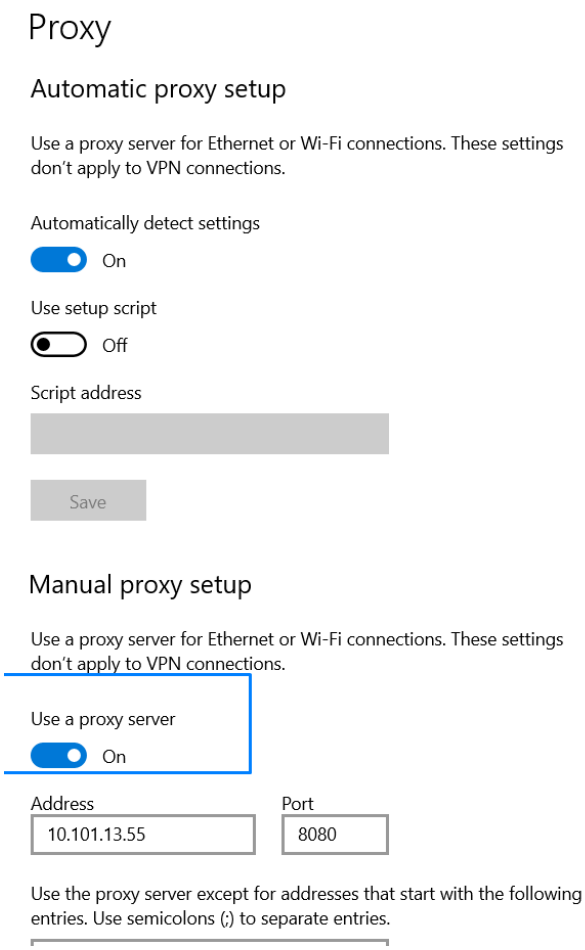
You should get a “successfully connected” message

You are now connected to the TELCO VPN and can access its respective websites. Links will be provided.

Step 3 (Proxy settings)

Now, while you have access to the VPN you have lost access to the rest of the internet, which is not good. You need to listen to music while you code or browse stack overflow on how to use the *map* function in Java, you name it. To regain access to the rest of the internet you need to configure your proxy settings. Press the Windows start button to search and look for Proxy Settings


All
Apps
Documents
Settings
Email
Web
More
Feed
Best match
Proxy settings
System settings
Settings
Change proxy settings
Change manual proxy server settings
Turn automatic proxy detection on or off
Use automatic proxy configuration
Search the web
Proxy - See web results
Proxy settings
System settings
Open
Proxy settings



You need to use a proxy server.

Here is the full configuration and here is the needed values for you to copy paste.

Address: XX.XXX.XX.XX

Port: 8080
Exclude: 10.*;*.xxx.gr*;*.xxx.gr*;*localhost*;127.0.0.1*

Once completed, make sure you click the checkbox and click Save. You now can reach anything TELCO connected while having a connection with the rest of the world.

The use of the VPN, however, leads to some security and lack-of-trust issues. To solve these issues, you need to import a couple of certificates. These certificates are: 1) /certificates_8/xxx.crt, 2)TELCO-issuing and 3) xxx.cer

If you feel like changes have not yet taken effect, please restart your machine, connect to the VPN, enable the proxy server, and try again.

Step 4 (JDK, IDE, GIT) *and many...many more abbreviations.*

You want to write microservices in Java. That is great. You need the Java Development Kit (JDK). You can find it here. You will need an oracle account. Please, use these credentials.

Username: xxx.xxx@xxx.com

Password: xxx

Now, you can write code, you just need a place to write it. What better place than IntelliJ. You can download the Ultimate version here. Follow the installation wizard. At some point, you will be asked for an activation code. Note, that the Ultimate version is **not** free but it is provided as such to us through the company. The activation code is shown below and you can copy paste it.

xxx

Great, that is two abbreviations down. You have the JDK and the IDE, but you do not have a project to write code for. You need to clone one. To clone one, you need GIT. Not really an abbreviation, but it sure looks like one. First download git and follow the installation guide. Use all the pre-defined installation options. Once git is installed restart your machine to apply all changes.

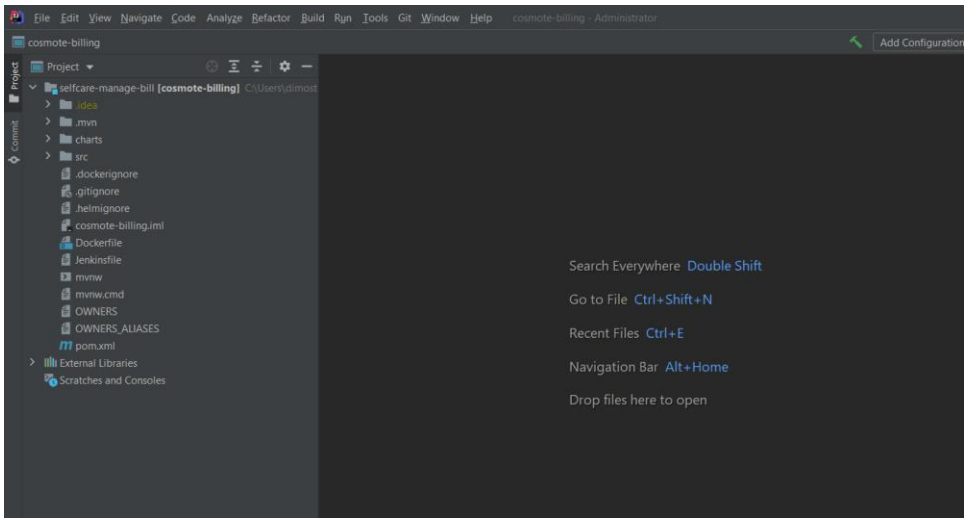
With git ready to go, you can clone a repository on your local machine to start working on it. You can access the full list of the project's repositories here. To connect use the following:

Username: XXX
Password: XXX

Do not get overwhelmed. The project is huge and ongoing. The µS suite is only a subset of those repositories. With time you will get to know and/or work on most of them. For now, look for xxx which is the latest at the time of the writing of this doc. Clone that to your local machine.

We suggest that you create a µS related folder inside the *C:\Users\your.name* directory. Inside that folder, create two folders. One named **xx** and one named **xxx**. None of these names are a hard requirement. You can name them per your request. Just know that you will need to clone all µS related projects inside websites while **xx** is for other use which will be shortly explained.

Once the project is cloned, fire up IntelliJ and open the project folder. You will be asked to "Trust the Project." Nothing suspicious going on. Trust the project. You should be greeted with the following page.



The first thing you would want to do is checkout to a different branch. Right now, you are on master. We normally work on *develop*. Switch to that. On the root directory of the repository, you should see a *pom.xml*. If you are unfamiliar with pom files, please take some time to google around and get acquainted. In short, you can think of it as a dependency management tool, like npm, yarn etc.

Once you get familiar with pom you will get to know what a parent is. The xxx µS, and every µS on the project has a parent. The xxx-ms-starter-spring-boot.

While the version might change with time, and maybe the version on your pom.xml is different than the one in the screenshot, the group and artifact id's will be the same. Pom does not know how to install or sync that project. Therefore, you need to manually download it. You can find it here.

The parent µS consists of several µS's as seen below.

Open the xxx-xx folder on your IntelliJ. If prompted, select to open it in a new window. In the root pom.xml file but also in all the pom.xml files of the rest of the children projects you should be able to see the following.

You need to make sure that the version is 1.2 in all of them. Please note that, in xx-ms-starter-spring-boot there is no such parent but you should be able to find this inside the pom.xml

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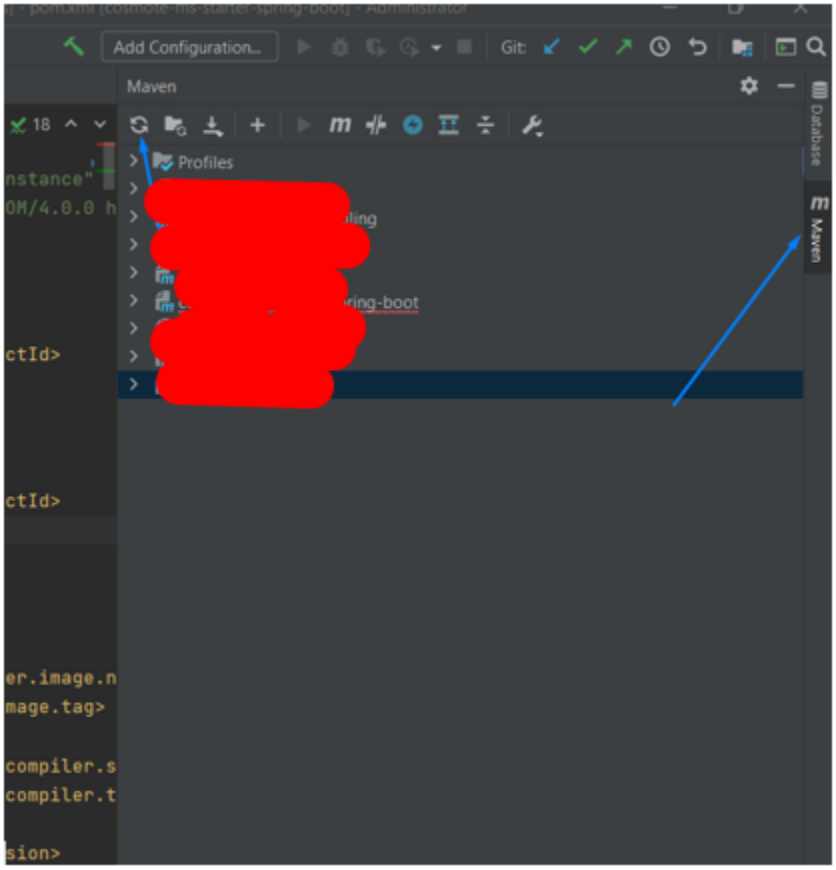
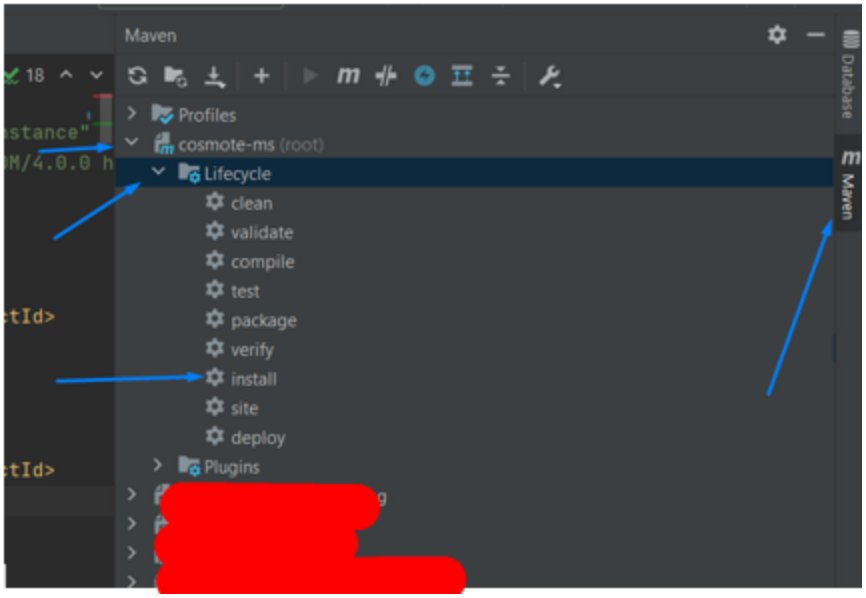
Make sure that the version is the same

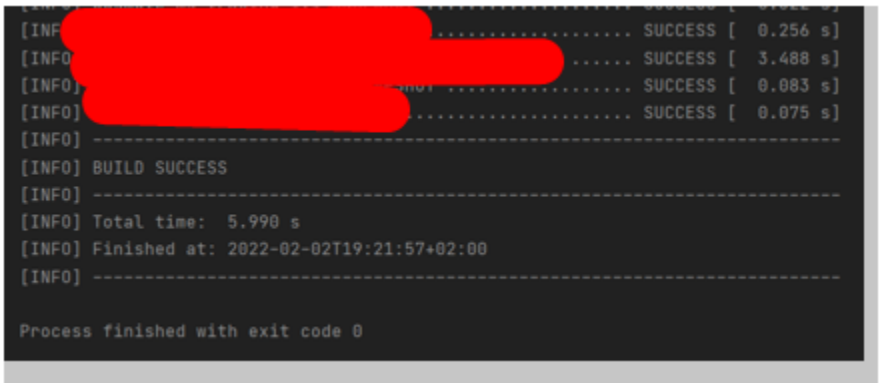
Once you make sure everything is aligned, and while being **disconnected** from the VPN, sync the root pom.xml file.



This might take a few seconds. Be patient and have a sip of coffee. Once ready, meaning you get no errors, you can perform a maven install for the root project.



This will fetch every dependency from the official maven repository. This will also take some time. Take a break. Walk around for a few minutes and get some off-screen time. OK, are you back? You should be able to see this in your terminal.



That is it, all dependencies are installed. You can start browsing the code or even adding to it. But you cannot really stage, commit, or yet alone push any code. Why is that? Git needs to be configured to work with the proxy/vpn combo we have set up. Create a .gitconfig file inside the *C:\Users\your.name.* Use IntelliJ to edit that file. Any text editor will do but come on... you are a developer.

You need to add the following configuration.

```
[user]
        name = firstName.lastName
        email = firstName.lastName@xxx.com
[alias]
        noproxy = !git config --global http.proxy ''  && git config --global https.proxy
```

```
''
        proxyote = !git config --global http.proxy http://xxx.xxx.gr:8080 && git config -
-global https.proxy https://xxx.xxx.gr:8080
        proxyvpn = !git config --global http.proxy http://xx.xxx.xx.xx:8080 && git config
--global https.proxy https://xx.xxx.xx.xx:8080

[http "https://xxx.xxx.gr/"]
    sslVerify = false
   proxy =
[https]
   proxy = https://xx.xxx.xx.xx:8080
```

Now, by using *git noproxy* and *git proxyvpn* you will be able to bounce between using git without and with a proxy respectively. Since the µS team works connected to the VPN and with the proxy settings enabled we work with git proxyvpn. Good news is that you only need to execute that command once. After that you can use git as you always use it without having to worry about anything else. If prompted for credentials, use the ones you used to connect to gitlab.

Step 5 (Enjoy µS)

You are finally done. It might have taken some time and a couple of cups of coffee but your environment is set up and our team cannot wait to work with you! If you have any questions, requests or faced issues and/or errors that were not mentioned in the doc, please, let us know so we can further improve the onboarding process.

Welcome, and code away!